Zheng Qing

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Education

Purdue University **Expected May 2027**

B.S. in Electrical Engineering | Cumulative GPA 3.83/4.0 | Major GPA 3.95/4.0 *West Lafayette, IN*

- **Relevant coursework:** Linear Control Systems, Signals & Systems, Probabilistic Methods, Digital System Design, Microprocessor Systems, Analog Circuits.

Research Experience

Senior Design — 3D Vehicle State Estimation and NMPC Trajectory Tracking **Sept 2026 – Present**

Advised by Prof. Ziran Wang, Purdue Digital Twin Lab *West Lafayette, IN*

- Porting the TUM FTM 3D vehicle dynamics state estimator — a 3D point-mass EKF supported by a single-track UKF — from a race-car platform onto a research vehicle with a weaker sensor set, in **C++** on **ROS 2 Humble**.
- Implementing an **NMPC** trajectory-tracking controller on top of the estimator so that the effect of estimation quality on closed-loop tracking error can be measured rather than assumed.
- Characterizing which state component — longitudinal velocity, yaw rate, or sideslip angle — dominates tracking degradation when GNSS drops out.

Purdue VIP ARES (Autonomous Race-Engineering Systems), Control Team **Jan 2026 – May 2026**

Undergraduate Researcher *West Lafayette, IN*

- Developed the low-level controller through a full progression — **PID** → **ADRC** → **LQR** → **MPC** — moving to optimal control once the objective had to trade tracking error, control effort, and stability against strategy-dependent priorities.
- Contributed to a hierarchical **strategy / planning / execution** architecture connecting high-level team objectives to real-time multi-agent control.
- Built a modular **ROS 2** evaluation pipeline — topic-level control, odometry state feedback, and time-synchronized logging — and ran comparative analysis of control strategies on a simulated Shanghai circuit.
- Co-authored the resulting team technical report and the companion conference poster.

Quantum Optics Laboratory **May 2025 – Oct 2025**

SURF Research Fellow, mentored by Prof. Shengwang Du | $6,500 fellowship *West Lafayette, IN*

- Designed and built a current-control module for a Magneto-Optical Trap end to end: **PCB design**, thermal management in a fan-cooled metal enclosure, and circuit validation with Hall-probe feedback.
- Presented a poster on biphoton generation and the MOT current-control box at the SURF Summer Symposium.

Poster Publications & Presentations

A Strategy-Driven Multi-Agent Control Framework for Autonomous Racing **Spring 2026**

- Z. Wang, K. Zhu, **Z. Qing**, K. V. Desireddi, J. Zhang, B. Lee. Purdue VIP ARES Control Team group technical report (not peer-reviewed).

Controller Design for an Autonomous Racing Vehicle — poster, CCAT Global Symposium on Mobility Innovation, University

- **ITS Michigan Scholar Award** — undergraduate winner, student poster competition.

Additional Experience & Projects

Undergraduate Teaching Assistant — Elmore Family School of ECE **May 2026 – Present**

Purdue University *West Lafayette, IN*

- **ECE 20875 — Python for Data Science** (Fall 2026): core EE course covering Python data structures and higher-order functions alongside probability distributions, sampling and hypothesis testing, regression with regularization and cross-validation, clustering, and classification.
- **ECE 20001 — Electrical Engineering Fundamentals I** (Summer 2026): supported a cohort of ∼**300 students** in Purdue's introductory circuits course — DC and AC circuit analysis, first-order transients, and diode and transistor fundamentals.

Purdue SoCET VIP — System-on-Chip Extension Technologies **Aug 2024 – May 2025**

Research Student, Circuit Design Team *West Lafayette, IN*

- SRAM and clock design: circuit design in **LTspice** and **Cadence Virtuoso**, SystemVerilog FSMs, **KiCad** PCB layout, and RISC-V embedded work in C with peripheral drivers.
- Presented at the Spring Undergraduate Research Symposium to an audience of over 100.

Flairmicro — Digital IC Design & Packaging **Aug 2023 – Sept 2023**
Engineering Intern *Fuzhou, China*

- Worked on Bluetooth communication modules for automotive applications; assisted with digital IC packaging, layout optimization, and production-line ESD compliance.

VGA Pong on FPGA *(SystemVerilog, Pico-ICE FPGA)* — ECE 270 team project **Mar 2025 – May 2025**

- Designed and verified the VGA timing generator (640×480 @ 60 Hz) and the pixel rendering module; implemented FSM-based game logic and collision detection through simulation, synthesis, and hardware debugging.

Honors — Honorable Mention, Mathematical Contest in Modeling (MCM/ICM); Semester Honors and Dean's List, Purdue Universi

Technical Skills

Programming/HDL: C++, Python, C, MATLAB, SystemVerilog
Robotics & Control: ROS 2 (Humble), EKF/UKF, NMPC, LQR, trajectory tracking, sensor fusion, bundle adjustment, OpenCV
EDA/Tools: Cadence Virtuoso, LTspice, KiCad μVision, Git, Docker, VS Code
Platforms/Systems: Linux
Spoken Languages: Mandarin (native), English (fluent)

Leadership & Extracurricular

PUCSSA — Chinese Students and Scholars Association at Purdue **Sept 2025 – Present**
Minister, Career Development *West Lafayette, IN*

- Organize career development events: resume workshops, mock interviews, and alumni networking sessions. Collaborate with companies like Midea, Eli Lilly, Huaqin

Music **Sept 2021 – Dec 2023**
President, High School Music Club; Lead Guitarist, One Tap Band *Suzhou, China*

- Lead guitarist in a local band performing British rock, hard rock, and punk metal; coordinated ensemble rehearsals and mentored junior members.